

GeoAir-Pulse

Open-source, full-stack GeoAI platform for hyper-local PM2.5 prediction and urban heat exposure mapping

Built on top of PostgreSQL, Python, ML, Mapbox, Deck.gl, Docker

THE CHALLENGE

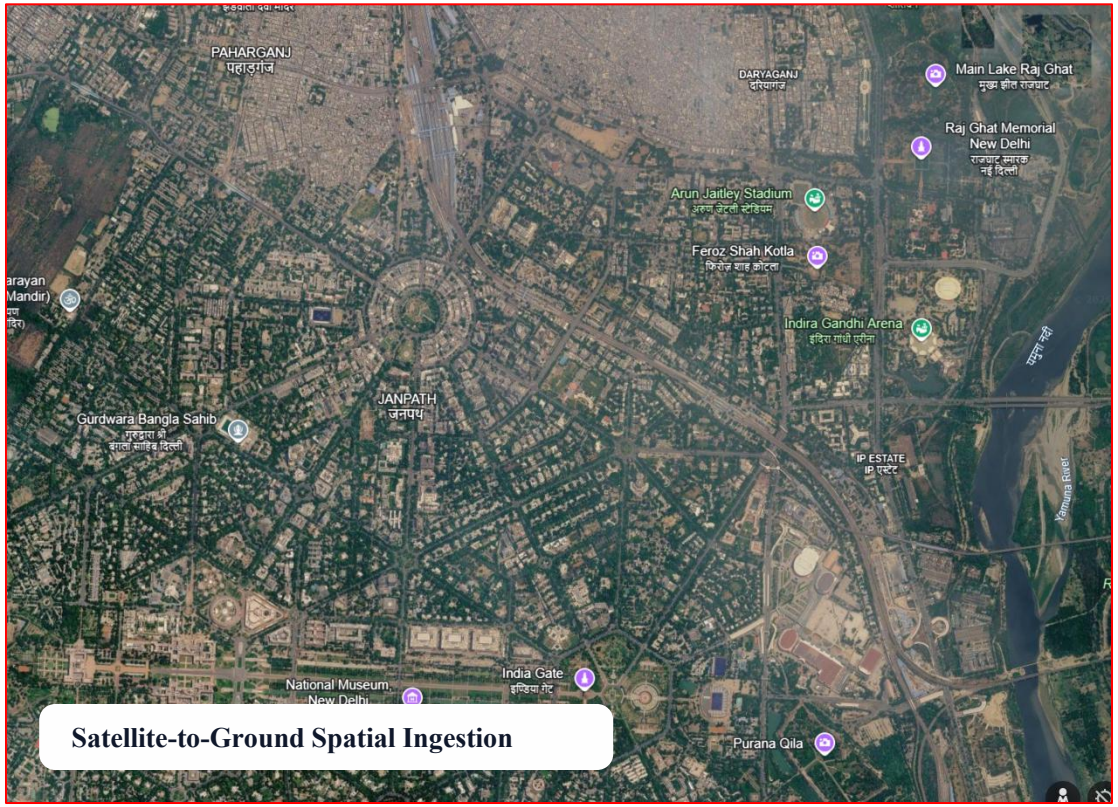
- Urban Air Quality Monitoring is too sparse
- Traditional monitoring in Delhi relies on expensive, widely dispersed ground stations, leaving critical spatial gaps in neighborhood-level exposure.

Few

Ground stations per district

High

Spatial variance in PM2.5



AIM & OBJECTIVES

Democratizing hyper-local environmental data.



The Aim

Build a 100% open-source, scalable GeoAI platform generating 100m-resolution PM2.5 and urban heat maps.



For Government

Enable targeted interventions, optimized sensor placement, and localized public health alerts.



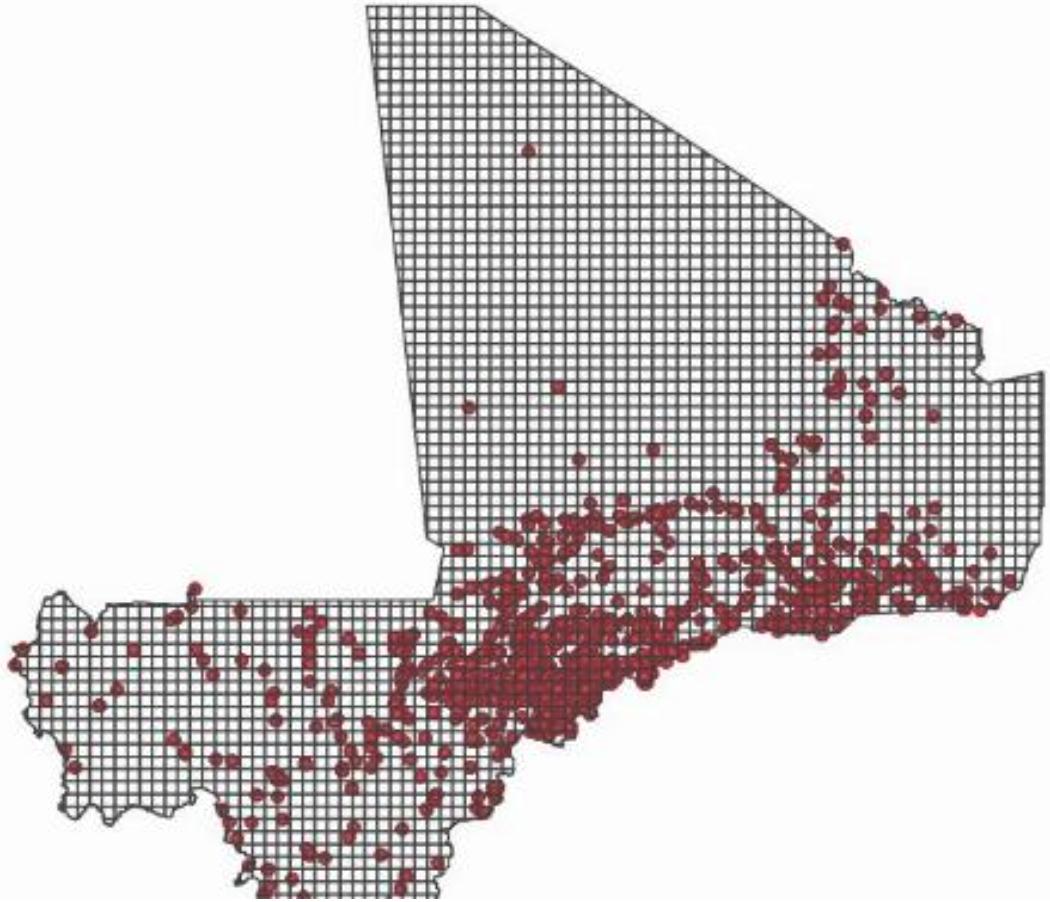
For Society

Provide equitable access to neighborhood-level exposure data, empowering vulnerable communities.

METHODOLOGY — STEP 1

Creating 100m Grids Across AOI

- Defining bounding box covering Delhi
- Building 100m spatial cells across the urban extent.
- Storing 345,912 candidate grid cells into PostgreSQL database.



(76.84 to 77.35 E, 28.40 to 28.88 N).

METHODOLOGY — STEP 2 & 3

Fetching Ground Truth & OSM Covariates

- Ground Stations: Ingesting real-time PM2.5 and PM10 sensor data from OpenAQ network across Delhi.
- Urban Form: Extracting road networks and building footprints from OpenStreetMap via Overpass API.



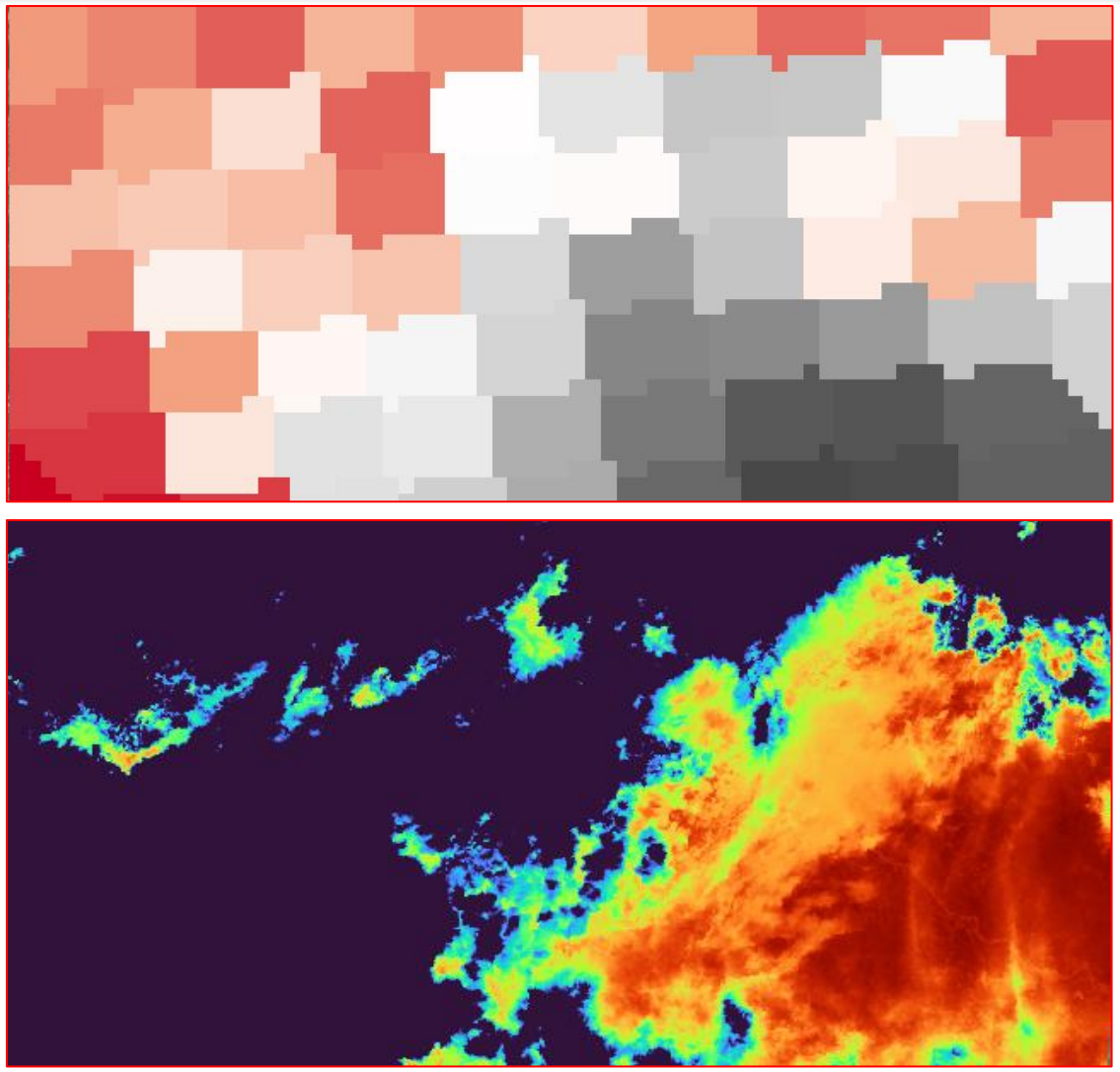
OSM building footprint and Road Network Ingestion

METHODOLOGY — STEP 4

Calculating Satellite Covariates

- Sentinel-5P NO2: Processing atmospheric column density rasters to capture pollution precursors.
- Landsat LST: Extracting Land Surface Temperature to map urban heat island effects.
- Zonal Sampling: Sampling raster values across all grid cell centroids using Rasterio.

Sentinel-5P NO2 – 1 KM resolution
Landsat LST – 30 metre resolution

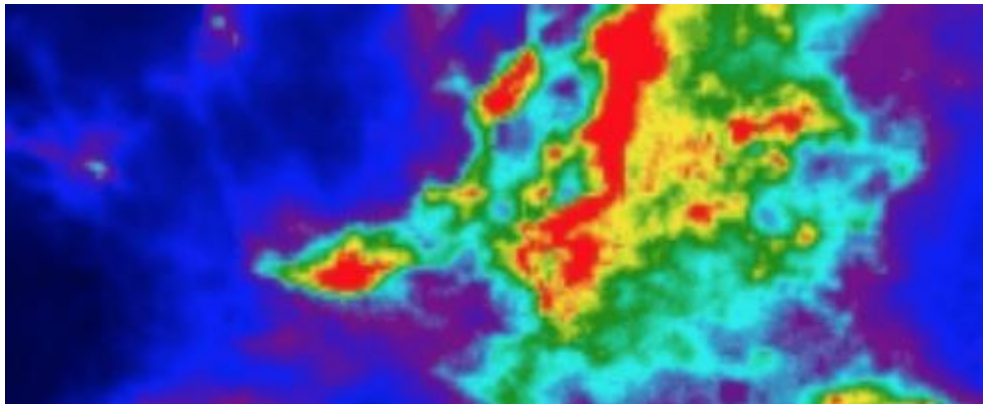


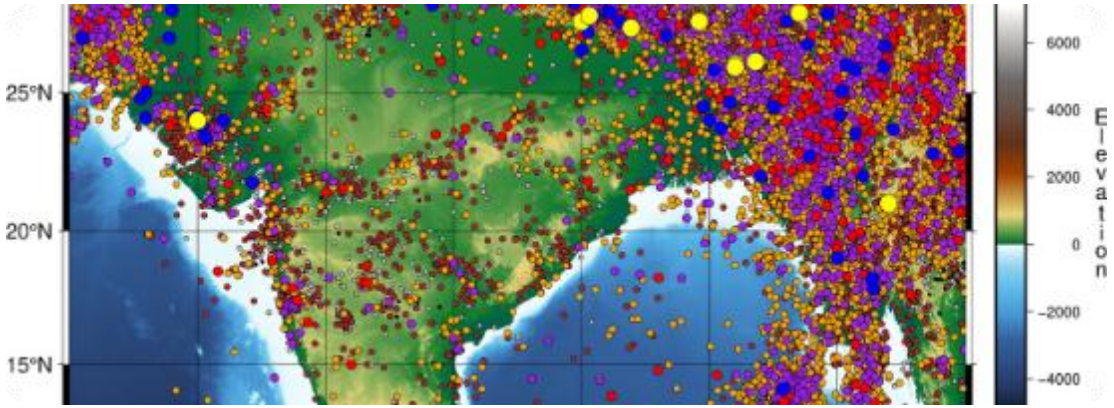
Sentinel-5P NO2 and Landsat LST raster data

GEOAI MODELING

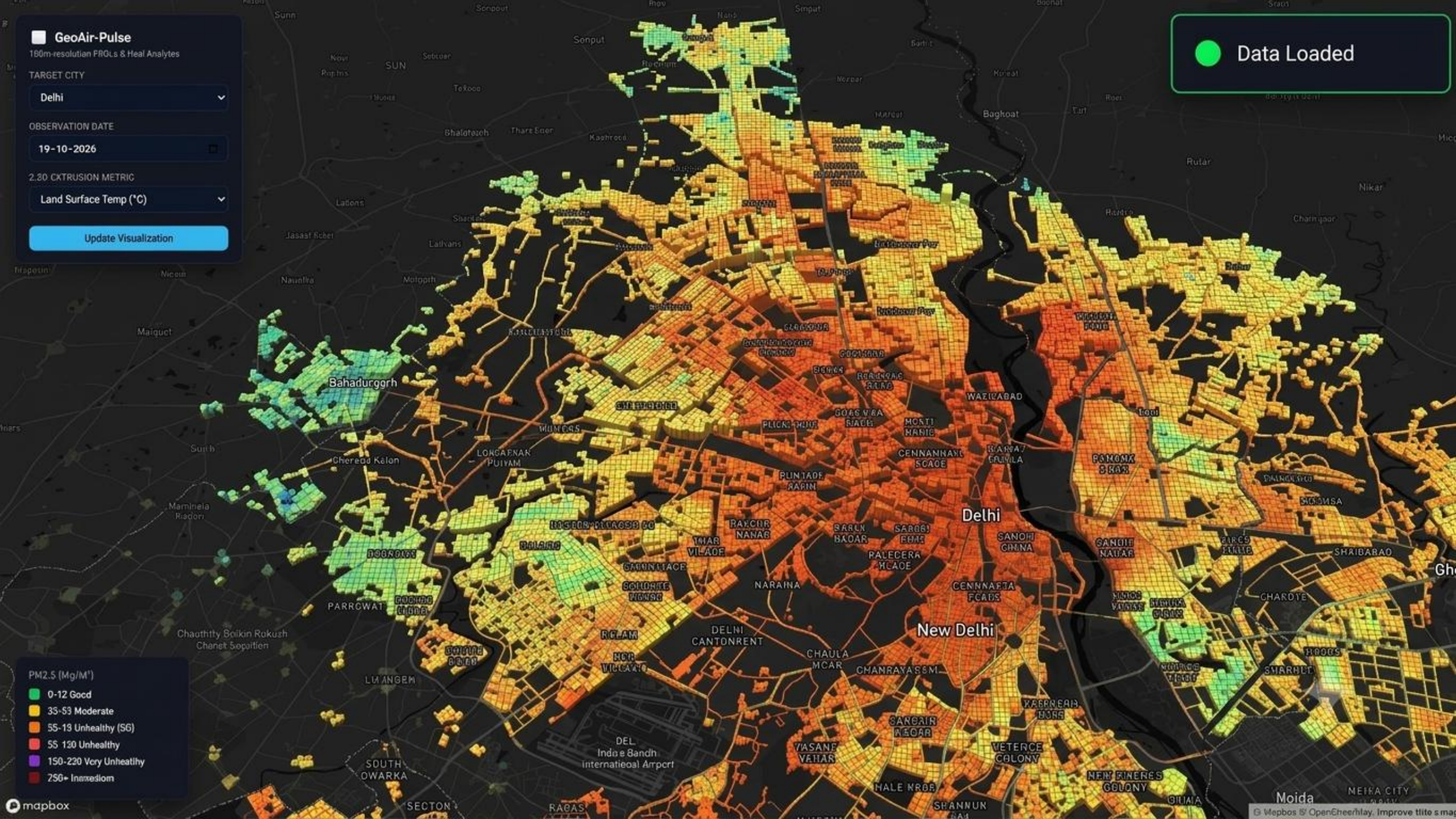
Model Training & Grid Predictions

- Random Forest Regressor: Training spatial ML models combining ground monitors with satellite covariates.
- Full-City Scoring: Generating localized predictions for all grid cells across Delhi.





localized predictions with ML inference


GeoAir-Pulse
160m-resolutian FRGLs & Heal Analytes
TARGET CITY
Delhi
OBSERVATION DATE
19-10-2026
2.30 CXTRUSION METRIC
Land Surface Temp (°C)
Update Visualization
Data Loaded
PM2.5 (Mg/M³)
0-12 Gocd
33-53 Moderate
55-13 Unhealthy (SG)
55 130 Unhealthy
150-230 Very Unheatihy
250+ Inmediom
Delhi
New Delhi
Bahadurgarh
Gherecd Kelon
PARROWAT
LIANGEM
SOUTH OWARKA
SECTON
DEL
Indo e Bandh
internatieoal Airpcrt
Moida
mapbox

Actionable intelligence for cleaner cities.

GeoAir-Pulse successfully bridges raw satellite rasters and sparse ground monitors into a unified, high-resolution geospatial intelligence pipeline.

GITHUB REPOSITORY

https://github.com/Prakash89192/GeoAirPulse-Prototype

Opensource Licensing

PostgreSQL: postgresql.org/about/licence
PostGIS: postgis.net/docs/manual-1.5/gpl.html
Python: docs.python.org/3/license.html
GeoPandas: github.com/geopandas/geopandas
Rasterio: github.com/rasterio/rasterio
GDAL / OGR: gdal.org/en/stable/license.html
Shapely: github.com/shapely/shapely
OSMnx: github.com/gboeing/osmnx
Scikit-Learn: github.com/scikit-learn/scikit-learn
XGBoost: github.com/dmlc/xgboost

FastAPI: github.com/fastapi/fastapi
SQLAlchemy: github.com/sqlalchemy/sqlalchemy
Uvicorn: github.com/encode/uvicorn
Deck.gl: github.com/visgl/deck.gl
Mapbox GL JS: mapbox.com/legal/tos
Docker: github.com/moby/moby
OpenAQ API: openaq.org
Copernicus Sentinel-5P: sentinels.copernicus.eu
Landsat 8/9 (USGS/NASA): usgs.gov
OpenStreetMap (OSM): openstreetmap.org/copyright